Glowlogics Internship Program

Python Assignment – 2

Introduction

This assignment is designed to strengthen your understanding of Python fundamentals by applying them to practical scenarios. Each question represents a simplified version of tasks performed in real software applications.

Before writing code, carefully analyze the problem, identify the required inputs, process the information logically, and then produce the expected output.

Remember:

Programming is not about writing code first. Programming is about understanding the problem and designing a solution.

Section 1: Variables and User Input

Question 1: Student Introduction System

Objective

Learn how to collect information from a user and store it in variables.

Problem Statement

Create a program that asks the user to enter:

- **Name**
- **Age**
- **College Name**
- **Department**

After collecting the details, display a personalized introduction message using an f-string.

Why This Matters

Most applications begin by collecting user information. Registration forms, admission portals, and profile creation pages all use similar logic.

Requirements

- **Use the `input()` function for all fields.**
- **Store each value in a separate variable.**
- **Use an f-string to display the output.**
- **Make the output readable and professional.**

Skills Practiced

- **Variables**
- **User Input**
- **String Formatting**
- **F-Strings**

Question 2: Internship Registration Form

Objective

Understand how online forms collect and process user information.

Problem Statement

Create a registration form that asks the user for:

- **Full Name**
- **Email Address**
- **Mobile Number**
- **City**

After collecting the information, display a registration confirmation message.

Example

Thank you Sai Ravi!

Your internship registration has been completed successfully.

Email: sairavi@gmail.com

City: Hyderabad

Why This Matters

Every online registration system follows a similar process:

1. **Collect information**
2. **Store information**
3. **Confirm successful registration**

Skills Practiced

- **Variables**
- **User Input**
- **F-Strings**

Section 2: Type Conversion

Question 3: Simple Calculator

Objective

Understand why type conversion is necessary when performing mathematical operations.

Problem Statement

Accept two numbers from the user and calculate:

- **Addition**
- **Subtraction**
- **Multiplication**
- **Division**

Display each result separately.

Important Note

The `input()` function returns data as a string.

To perform calculations, convert the inputs using:

- `int()`
- **or** `float()`

Real-World Relevance

Calculator applications, banking software, billing systems, and accounting platforms perform similar operations.

Skills Practiced

- **Type Conversion**
- **Arithmetic Operators**
- **User Input**

Question 4: Age After 10 Years

Objective

Learn how numerical data can be manipulated after converting user input.

Problem Statement

Ask the user to enter their current age.

Calculate and display what their age will be after 10 years.

Example

Current Age: 20

Output:

Your age after 10 years will be 30.

Skills Practiced

- **Integer Conversion**
- **Mathematical Operations**

- **Variables**

Section 3: Lists

Question 5: Favorite Movies Collection

Objective

Learn how lists store multiple values in a single variable.

Problem Statement

Create a list containing five favorite movies.

Perform the following operations:

1. **Print the complete list.**
2. **Print the first movie.**
3. **Print the last movie.**
4. **Print the total number of movies.**

Why This Matters

Streaming platforms like Netflix and Amazon Prime store movie collections using data structures similar to lists.

Skills Practiced

- **Lists**
- **Indexing**
- **len() Function**

Question 6: Student Welcome System

Objective

Understand how loops help process multiple records efficiently.

Problem Statement

Create a list containing ten student names.

Use a loop to display a welcome message for each student.

Example

Welcome Sai

Welcome Hasan

Welcome Priya

Real-World Relevance

Email systems often send welcome messages to hundreds or thousands of users automatically.

Skills Practiced

- **Lists**
- **Loops**
- **Iteration**

Question 7: Attendance Register

Objective

Learn how to combine loops and list indexing.

Problem Statement

Create a list of students and display their names with serial numbers.

Example

1 Sai

2 Abhi

3 Hasan

4 Priya

Real-World Relevance

Attendance systems, employee records, and customer databases use numbering systems to organize information.

Skills Practiced

- **Lists**
- **Range Function**
- **Loops**

Section 4: Conditional Statements

Question 8: Voting Eligibility Checker

Objective

Understand decision-making using conditional statements.

Problem Statement

Ask the user for their age.

If the age is 18 or above, display:

You are eligible to vote.

Otherwise display:

You are not eligible to vote.

Real-World Relevance

Government portals and verification systems use age-based eligibility checks.

Skills Practiced

- **Comparison Operators**
- **if Statements**

- **Logical Decision Making**

Question 9: Grade Calculator

Objective

Learn how multiple conditions can be evaluated using `if-elif-else`.

Problem Statement

Accept marks from the user and assign grades according to the grading criteria.

Requirements

90 and above → A Grade

70–89 → B Grade

50–69 → C Grade

Below 50 → D Grade

Real-World Relevance

School management systems and examination portals automatically calculate grades using similar logic.

Skills Practiced

- **if**
- **elif**
- **else**
- **Comparison Operators**

Question 10: Login Verification System

Objective

Understand basic authentication concepts.

Problem Statement

Store a username:

admin

Ask the user to enter a username.

If the entered username matches the stored username:

Login Successful

Otherwise:

Invalid Username

Real-World Relevance

Every website login page uses validation logic before granting access.

Skills Practiced

- **Variables**
- **Conditions**
- **String Comparison**

Section 5: Loops

Question 11: Print Your Name 20 Times

Objective

Understand how loops automate repetitive tasks.

Problem Statement

Print your name 20 times using a `for` loop.

Why This Matters

Instead of writing the same statement repeatedly, programmers use loops to automate repetition.

Skills Practiced

- **For Loops**
- **Range Function**

Question 12: Even Number Generator

Objective

Learn how conditions can be combined with loops.

Problem Statement

Print all even numbers between 1 and 50.

Challenge

Try finding two different methods to solve this problem.

Skills Practiced

- **Loops**
- **Modulus Operator**
- **Conditions**

Question 13: Multiplication Table Generator

Objective

Generate repetitive mathematical output using loops.

Problem Statement

Accept a number from the user and display its multiplication table from 1 to 10.

Real-World Relevance

Educational applications frequently generate tables dynamically.

Skills Practiced

- **Loops**
- **User Input**
- **Arithmetic Operations**

Question 14: Password Verification System

Objective

Learn how programs can continue running until a condition is satisfied.

Problem Statement

Keep asking the user to enter a password until the correct password is entered.

Correct Password:

1234

Real-World Relevance

Login systems repeatedly request credentials until valid information is provided.

Skills Practiced

- **While Loops**
- **Conditions**
- **Repetition Logic**

Section 6: F-Strings

Question 15: Personalized Greeting System

Objective

Learn how dynamic messages are generated using user data.

Problem Statement

Ask the user for:

- **Name**
- **Course**

Display a personalized greeting.

Example

Hello Sai Ravi, welcome to the Python Course.

Skills Practiced

- **User Input**
- **F-Strings**

Question 16: Shopping Bill Generator

Objective

Create a simple billing system.

Problem Statement

Collect:

- **Product Name**
- **Quantity**
- **Price**

Calculate the total cost and display a formatted bill.

Bonus Challenge

Add GST or discount calculations.

Real-World Relevance

Retail billing software and e-commerce applications perform similar calculations.

Skills Practiced

- **Variables**
- **Type Conversion**
- **F-Strings**

Section 7: Dictionaries

Question 17: Student Biodata

Objective

Learn how dictionaries store related information using key-value pairs.

Problem Statement

Create a dictionary containing student information.

Display the complete dictionary.

Skills Practiced

- **Dictionaries**
- **Data Organization**

Question 18: Access Dictionary Values

Objective

Learn how to retrieve information from dictionaries.

Problem Statement

Print Name, College, and Branch separately using dictionary keys.

Skills Practiced

- **Dictionary Access**
- **Key-Value Retrieval**

Question 19: Update Information

Objective

Understand how existing data can be modified.

Problem Statement

Change the city value and display the updated dictionary.

Real-World Relevance

Users frequently update profile information in applications.

Question 20: Add Additional Information

Objective

Learn how to expand dictionaries dynamically.

Problem Statement

Add:

- **Email**
- **Phone Number**

Display the updated dictionary.

Real-World Relevance

Applications continuously collect and store additional user information.

Question 21: Digital ID Card Generator

Objective

Combine dictionaries and f-strings to create a professional information display.

Problem Statement

Create a digital ID card using dictionary data.

Display the information neatly using f-strings.

Real-World Relevance

College portals, HR systems, and internship management platforms generate digital ID cards similarly.

Mini Project 1: Amazon User Profile

Objective

Create a simplified user profile similar to an e-commerce platform.

Expected Features

Store:

- **Name**
- **Email**
- **Orders**
- **Prime Membership Status**
- **City**

Display a personalized dashboard message.

Learning Outcome

Understanding how user profiles are stored and displayed.

Mini Project 2: Student Management System

Objective

Build a simple student profile management system.

Expected Features

- **Collect student information**
- **Store data in a dictionary**
- **Display the information professionally**

Learning Outcome

Understanding how management systems organize user data.

Bonus Challenge

Student Database System

Create a list containing five student dictionaries.

Each student should contain:

- **Name**
- **Age**
- **College**
- **Department**

Using a loop:

Display the names of all students.

Advanced Challenge

Display complete information for every student.

Real-World Relevance

This is a simplified version of how schools, colleges, and companies manage large datasets.

Submission Guidelines

1. **Write clean and readable code.**
2. **Follow proper indentation.**
3. **Use meaningful variable names.**
4. **Add comments wherever necessary.**
5. **Test all programs before submission.**
6. **Try solving independently before seeking help.**

Final Note

The purpose of this assignment is not only to learn Python syntax but also to develop logical thinking and problem-solving abilities.

The more you practice, the stronger your programming foundation becomes.

Good luck and happy coding!

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